

Predicting Factor Decay: ML Models for Cross-Factor Predictability Erosion

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A financial relationship can help a forecasting model for years and then weaken. This paper asks whether the change in that relationship can itself be anticipated.

The research question

A factor is a return series representing an investment characteristic, such as value, size, or momentum. The motivating example asks whether past value-factor returns (HML) help predict the size factor (SMB), and how long that predictive contribution lasts. The focus is statistical predictability and its durability, rather than the economic cause of a factor's return.

How the study works

The primary analysis covers six daily factors over 1990-2024 and 30 directed pairs. A Student-t hidden Markov model groups observations into statistical regimes. Within those regimes, Granger tests examine whether source-factor lags add information beyond the target's own history. Features such as initial signal strength and rolling variability then feed classifiers and survival models. The operational decay label concerns a loss of statistical significance within five years.

What the paper reports

A random survival forest achieves a five-fold cross-validation C-index of 0.933, with a fold standard deviation of 0.097. That metric measures ordering of event times, not return prediction accuracy. The analysis contains only seven decay events among thirty pairs. On a separate ten-pair holdout, classification accuracy is 60%, or six correct predictions. The main US value-to-size relationship fails corrected tests in the 2013-2024 evaluation. Some regional crisis-state effects remain, but they do not independently validate the decay model.

What the findings mean

The results motivate studying signal durability as a monitoring problem. They do not yet establish reliable future-time performance. Factor pairs share underlying series, the focal pair was selected after inspection, and some pre-2012 decay fits use regime assignments estimated with the full sample. The manuscript also describes the state-selection window inconsistently. A fully prospective evaluation would refit every stage using only data available at each training date. The separate 6.6-day rolling-HMM result measures regime detection delay after a change, not advance warning of factor decay.

KEY TAKEAWAY

The distinct question is when an observed predictive relationship weakens. The available evidence supports further exploratory testing, with uncertainty and information timing central to the next study.